



CSE 478 : Thesis Report

Osteoporosis Diagnosis Using X-ray Images

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ABSTRACT

Osteoporosis degrades the quality of bones and is the primary cause of fractures in the elderly and women after menopause. Generally for diagnosis of osteoporosis checked by scanning the density of bones. X-ray is the cheapest and most common imaging technique to check bone density. To measure bone mineral density (BMD) dual-energy x-ray absorptiometry used. Deep learning systems have gained the popularity in image analysis field over the last few decades. A convolution neural network (CNN) based approach to detect osteoporosis from x-rays. Using AlexNet, VggNet-16, ResNet, and VggNet19 transfer learning model based on Deep Learning x-ray images of knee joints classified into normal, osteopenia and osteoporosis groups. The results of the study suggest that a deep learning system with transfer learning can help clinicians to detect osteoporosis in its early stages hence reducing the risk of fractures.

Keywords: DeepLearning, Neural Network, CNN

STATEMENT OF ORIGINALITY

It is hereby declared that the contents of this thesis is original and any part it has not been submitted elsewhere for the award of any degree or diploma.

Signature of the Candidate
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Date : _____

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Chapter 1

INTRODUCTION AND OVERVIEW

Introduction

X-ray imaging is the most common imaging technique amongst the medical community to find bone pathologies. X-rays are the oldest and most common techniques to take images of almost all bones of the body like wrist, knee, elbow, shoulder, knee, pelvis, spine, etc. X-ray imaging helps in fracture diagnosis, dislocation of joints, bone injury, abnormal bone growth, infection. That is due to the weakening of bones caused by osteoporosis, cancer, or osteogenesis.

Medically osteoporosis is diagnosed with Dual Energy X-ray Absorptiometry Technique (DXA). which determines the bone mineral density (BMD) in terms of the T-score and Z-score values approved by WHO for different stages of osteoporosis. But it suffers from some limitations which include areal measurements with the technique being costly and less available. Other imaging modalities that are used for osteoporosis detection are the Quantitative Ultrasound System (QUS), Computed Tomography (CT), Magnetic Resonance Imaging (MRI). MRI is a 3 T improved bone microarchitecture imaging technology but is very costly and has lower spatial resolution, CT is 3D geometric imaging with volumetric measurements but has a high dose of radiation and doesn't qualify for the WHO's definition of osteoporosis detection. QUS is simple, non-invasive, portable, cost-effective, and uses sound waves for studying bones but it is site-specific and has an absence of strong empirical evidence. Considering these limitations, a cost-effective, readily available, and accurate detection system is required. This led the researchers

to take advantage of recent advancements in the field of imaging technology to analyze medical images with computer algorithms to form computer-aided diagnostic systems (CAD).

In recent years, among the CAD systems for medical image analysis, deep learning-based convolutional neural network (CNN) techniques have gained popularity due to their state-of-the-art results in detecting many diseases from images like brain tumor detection, breast cancer detection, pneumonia detection, cancer detection, human activity recognition, multiple sclerosis etc.

Many CAD systems are proposed for osteoporosis diagnosis including deep learning at various bone sites like hip, spine, hand, and tooth but not much work has been done to detect knee osteoporosis.

Here, we have used the power of CNN architectures and the cost-effectiveness of X-ray imaging to find the early detection system for knee osteoporosis. Our model uses the prominent CNNs namely AlexNet, VggNet-16, ResNet, and VggNet-19 for classifying the knee X-ray Images.

Chapter 2

RELATED WORK

Machine learning approaches especially deep convolution neural networks have shown state-of-the-art results in disease detection. Many researchers have successfully used machine learning approaches to build the osteoporosis diagnosis system from different types of images. In this section, we have discussed the latest works done in the field of osteoporosis diagnosis using deep convolution neural networks.

The magnetic resonance images of the proximal femur were studied by Deniz et al. [1] for osteoporosis detection. They used DCNN to segment the proximal femur for measuring the quality of bone and assessment of fracture.

Two CNN models namely MS-Net (Mark- Segmentation- Network) and BCC-Net (Bone- Conditions -Classification Network) were proposed by Tang et al. [2] for ROI selection and for bone type determination on basis of extracted features from ROI respectively in osteoporosis diagnosis and achieved 76.65

Liu et al. [3] diagnosed osteoporosis from x-ray images of the pelvis. They calculated the energy function from the softmax of the proposed U-net model that uses the deep features of the medullary joint from X-rays to detect osteoporosis. This study poorly diagnoses the images of the bone mass reduction group and osteoporosis group. Yamamoto et al. [4] detected osteoporosis from hip radiographs using CNN. They combined the clinical covariates with images and found that it improved the performance and the best performance was achieved by EfficientNet CNN.

The AlexNet Classifier was used by Tecle et al. for diagnoses of osteoporosis [5]. They used the

X-ray images of the hand and classified the osteoporotic and non-osteoporotic images from the segmented second metacarpal region.

He et al. [6] analysed the knee X-rays and proposed to use two radiographic parameters namely cortical bone thickness and distal femoral cortex for bone quality assessment. These parameters were found to have a significant correlation with BMD and T-score.

From the above-related works of literature, we could find that knee osteoporosis is an understudied field as compared to other sites like vertebrae and teeth. Detecting osteoporosis from the knee can protect vital organs like kidney, pancreas, etc. from getting exposed to harmful radiation while getting the images for analysis. X-rays are also the cheapest form of medical images available and can help build a cost-effective system. we have used the knee x-ray images which are classified as normal, osteopenic, and osteoporotic on basis of T-score values obtained from the QUS system to train the CNN networks. The transfer learning helps the CNN to perform well even when trained on a small dataset.

Chapter 3

MATERIALS AND METHODS

The knee x-ray images were collected from the BMD camp organized by the Unani and Panchkarma Hospital, Srinagar, J&K, India, and its sister branches from 21-12-2019 to 31-12-2019 in central Kashmir, north Kashmir, and south Kashmir. The camp was organized in the hospital premises and was open to participants from all age groups, genders, and different regions of Kashmir, India. The dataset consists of both x-ray images as well as osteoporosis-related clinical factors for each participant. Each patient first went through a personal interview wherein he was informed about the procedure of the QUS BMD test and various clinical factors like age, gender, height, previous history of fracture or any other pathology, lifestyle habits, medications, etc. were documented.

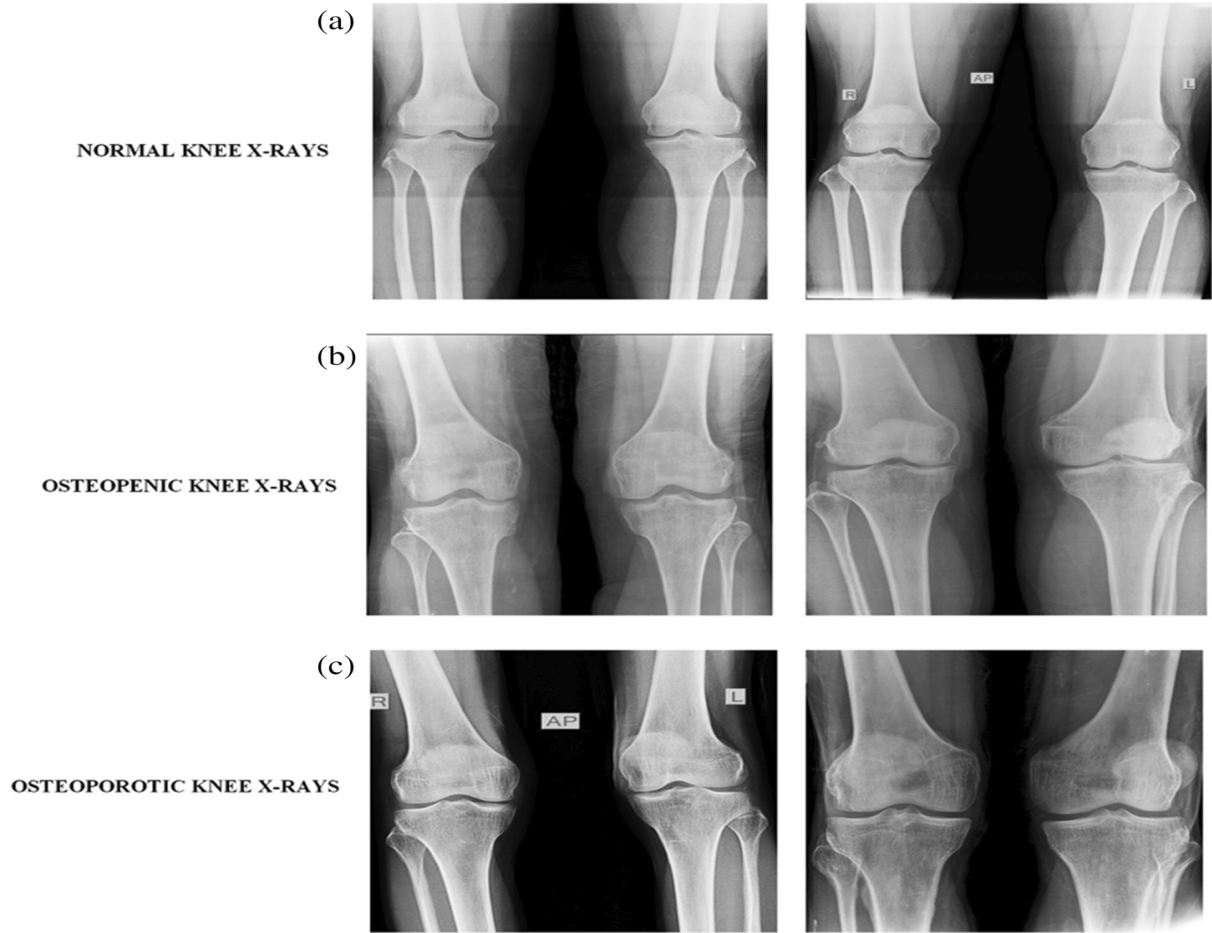


Figure 3.1: KNEE x-ray

In the x-rays collected from the camp, some X-rays had scans of both the knees so, the left and right knee x-rays were separated and then the dimensions of all the x-rays were kept the same, and finally, we have 381 knee x-rays.

Data Sets:

- Kaggle[7]
- Mendelye[8]

3.1 Proposed methodology

Figure 3.2 shows the block diagram of proposed model for detection of osteoporosis from knee x-rays. Firstly, the knee X-rays were collected as mentioned in Section 3.1 to form an image dataset which is then split into training data (used to train CNN classifier) and test data (used to test the trained classifier). The training data is augmented to increase the number of

images in the training set as CNN works better with more data. Then the imageset is passed on to CNN model for training the CNN. Then finally the prediction ratio of train and test data is analysed and performance of classifier to classify images into normal, osteopenia, and osteoporotic images is evaluated.

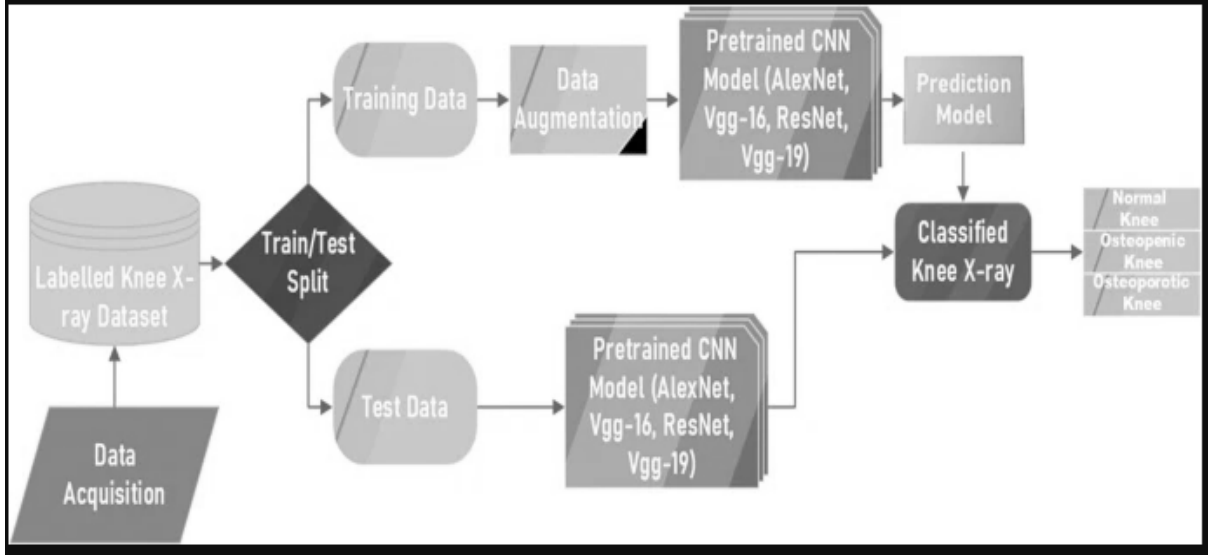


Figure 3.2: Proposed Methodology

Then finally the prediction ratio of train and test data is analysed and performance of classifier to classify images into normal, osteopenia, and osteoporotic images is evaluated.

3.2 CNN architectures

CNN is the variant of deep neural networks whose intermediate levels are based on the principle of convolution. The convolution is the mathematical function in which one function is modified with another function to get a new function with some modified features. CNNs are used for the processing of images in which the image is convolved with a filter of less length * width to reduce the size of the images but maintain the basic information contained in the image. CNN as compared to other deep learning architectures have received more interest from researchers because they can utilize both the configural and the spatial information of the 2D as well as 3D images. Many CNNs have been developed to solve various types of problems and they vary with each other in one or the other aspect but the basic components are the same. The CNNs consist of three types of layers viz.; convolutional layer, pooling layer, and fully connected layer. The convolutional layer is responsible for learning the feature representations of the

input images by using the set of filters. The pooling layer helps in reducing the computations and parameters with the downsampling of the representations to achieve the shift-invariance. It is usually placed in between the two convolutional layers. There could be any number of convolutional and pooling layers in the network. By stacking them properly we can extract the feature maps containing the higher-level representations. One or more fully connected (FC) layers are present at the end of the stacked convolutional and pooling layers and before the output layer, to perform the task of reasoning.

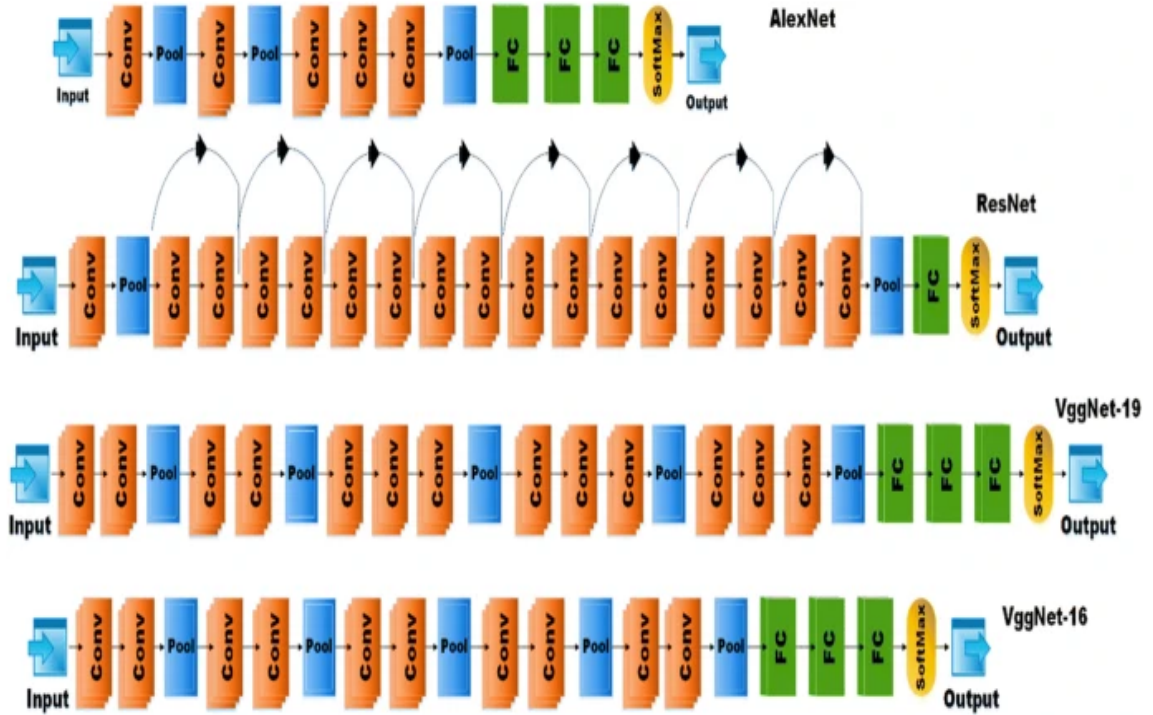


Figure 3.3: CNN architectures

Popular pretrained CNN architectures namely AlexNet, ResNet-18, VggNet-16, and VggNet-19.

Chapter 4

EXPERIMENTATION AND RESULTS

For experimentation, we have compared the performance of four CNN architectures namely AlexNet, ResNet 18, VggNet-16, and VggNet-19 for classifying the three stages of osteoporosis in knee x-rays. These architectures have been successfully used in classifying the medical images of other diseases so can be used for osteoporosis detection from knee x-rays. These architectures classify the images by extracting the feature maps of what is in a knee x-ray. These architectures vary from each other in way of the number of layers (convolution, pooling, or FCC), or some other units.

Chapter 5

DISCUSSION

This is the first study that is aimed to detect osteoporosis from knee X-rays that are classified into disease groups (osteopenia and osteoporosis) and normal on basis of BMD values obtained from the medical diagnostic test QUS. We have used the power of CNN networks to classify the class of X-ray images by interpreting the differences in the image groups and then classifying them automatically. The performance of well-known CNNs was compared in order to get the best performing CNN for detecting osteoporosis from knee X-rays. The participants of the study included participants of all genders and ages. Deep learning architectures have been used to detect osteoporosis from other sites like hand, spine, or hip scans.

Chapter 6

CONCLUSION

evaluated and compared the performance of popular CNN architectures namely ResNet-18, VggNet-16, AlexNet, and VggNet-19 in diagnosing osteoporosis from knee X-ray images. The X-ray images used were taken from the custom dataset that was classified into normal, osteopenia, and osteoporosis group with the help of a medically accepted BMD test known as the Quantitative Ultrasound system which calculates the T-score by measuring the BMD of bone. The custom dataset contained a total of 381 knee x-ray scans.

In the future, more data can be collected especially from normal and osteoporotic subjects. Secondly, we can find the relationship of knee osteoporosis with osteoporosis at other sites to make universal diagnostic system of osteoporosis. Thirdly, the system can be build which will detect osteoporosis from clinical factors in combination with images.

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